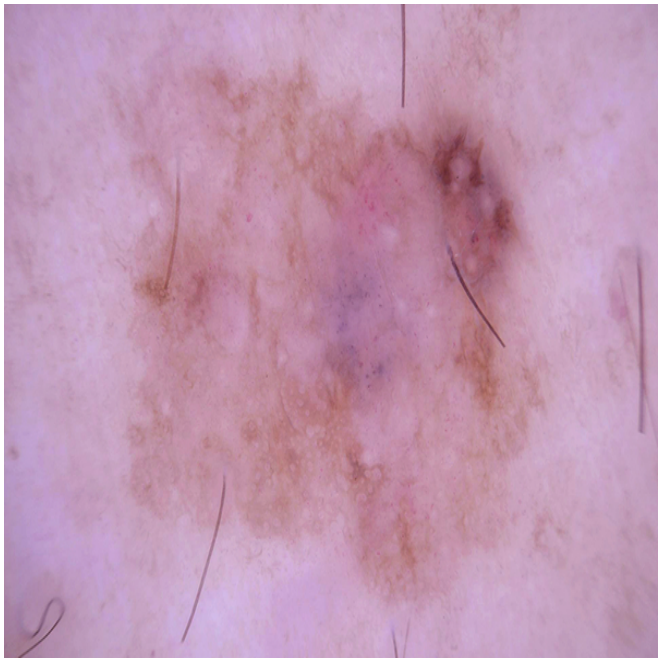
SkinGuard AI

Explainable AI Based Dermoscopic Skin Analysis Report

Scan Information

Scan ID	SG-20260924190551
Date & Time	24-09-2026 19:05
Predicted Disease	Melanocytic Nevus
Confidence	23.19 %
Severity	Low
Severity Score	0.0

1. Uploaded Dermoscopic Image



2. Image Quality Analysis

Parameter	Result
Quality Score	75 %
Blur	Low
Brightness	Good
Noise	Low
Overall Status	Good Image

3. Image Preprocessing

Original Image → Resized → Enhanced → Normalized

Status: Completed

4. Lesion Segmentation

Segmentation Model: U-Net

Status: Completed



5. Classification / Disease Detection

Parameter	Result
Predicted Class	Melanocytic Nevus
Confidence	23.19 %
Classification Model	EfficientNetB3

6. Seven-Class Probability Distribution

Disease Class	Probability
Actinic Keratosis	0.0 %
Basal Cell Carcinoma	0.0 %
Benign Keratosis	0.0 %
Dermatofibroma	0.0 %
Melanoma	0.0 %
Melanocytic Nevus	0.0 %
Vascular Lesion	0.0 %

7. Explainable AI - Grad-CAM

Grad-CAM highlights image regions that contributed to the model prediction.

Original Image



8. Severity Prediction

Severity Level	Low
Severity Score	0.0
Algorithm	Rule-Based Severity Prediction

9. Personalized Clinical Recommendation

Low severity has been detected. Continue regular skin monitoring and consult a dermatologist if any unusual changes are observed.

10. AI System Summary

Module	Technology
Image Preprocessing	OpenCV
Lesion Segmentation	U-Net
Disease Classification	EfficientNetB3
Explainable AI	Grad-CAM
Severity Prediction	Rule-Based Algorithm
Web Application	Flask

Medical Disclaimer: This AI-generated report is intended for screening and clinical decision support purposes only. It should not be considered a substitute for professional medical diagnosis. Please consult a qualified dermatologist for final evaluation.